



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE, MECH and CIVIL

Department of Information Technology Academic Year 2025 - 2026 (Even Semester)

Semester, Degree & Branch: II Semester, B.Tech & IT

Course Code & Title: IT25201& Foundations of Data Science using Python

Name of the Faculty member (s): Mrs. P.Ramya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Module I / Python Language Basics
- **Course Outcome:** CO1
- **Topic Learning Outcome:** -
- **Activity Chosen:** Crossword Puzzle
- **Justification:**

Using crossword puzzles to teach Python basics is an innovative practice because it transforms passive learning into a high-engagement, **active recall** exercise that reinforces technical syntax and keywords through gamification. This method is justified as it enhances **vocabulary retention** of specific terms like "boolean" or "tuple," improves **logical thinking and pattern recognition** essential for debugging, and provides a low-stress environment for students to self-evaluate their knowledge gaps in real-time. By moving beyond traditional rote memorization, puzzles foster a competitive yet collaborative atmosphere that significantly boosts student motivation and peer interaction.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - The time allotted for the topic is 10 Minutes.
 - Following the completion of Module 1, a 10-minute revision activity was conducted where students collaborated in groups to solve a 10-question paper-based crossword puzzle.
 - This exercise required teams to engage in critical thinking to identify the correct Python concepts, with sample results documented in figures 1a and 1b.
 - To conclude the session, all solutions were reviewed collectively to ensure every student achieved a full understanding of the module's core terminology.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO11	PSO1
CO1	3	3	3	3	3	2	3

(1 – Low 2 – Moderate 3 – High)

• **PO/PSO Mapped:**

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO11	PSO1
	3	3	3	3	3	2	3
Justification for Correlation	The students will apply the basics mathematics, science and engineering principles for solving problems using basic python Constructs.	Students will be able to handle the memory management using basic problem solving techniques.	Students will be able to provide solution to the real time problems by using necessary basic constructs in Python.	Students will be able to investigate the problems and provide valid conclusion using python problem solving techniques	Students will be able to use jupyter Notebook IDE to provide solution to the problems.	The students will improve their knowledge in python continuously by implementing in emerging technologies	The students will have ability to implement programs of varying complexity for web, cloud, or data science applications

• **Images / Screenshot of the practice:**

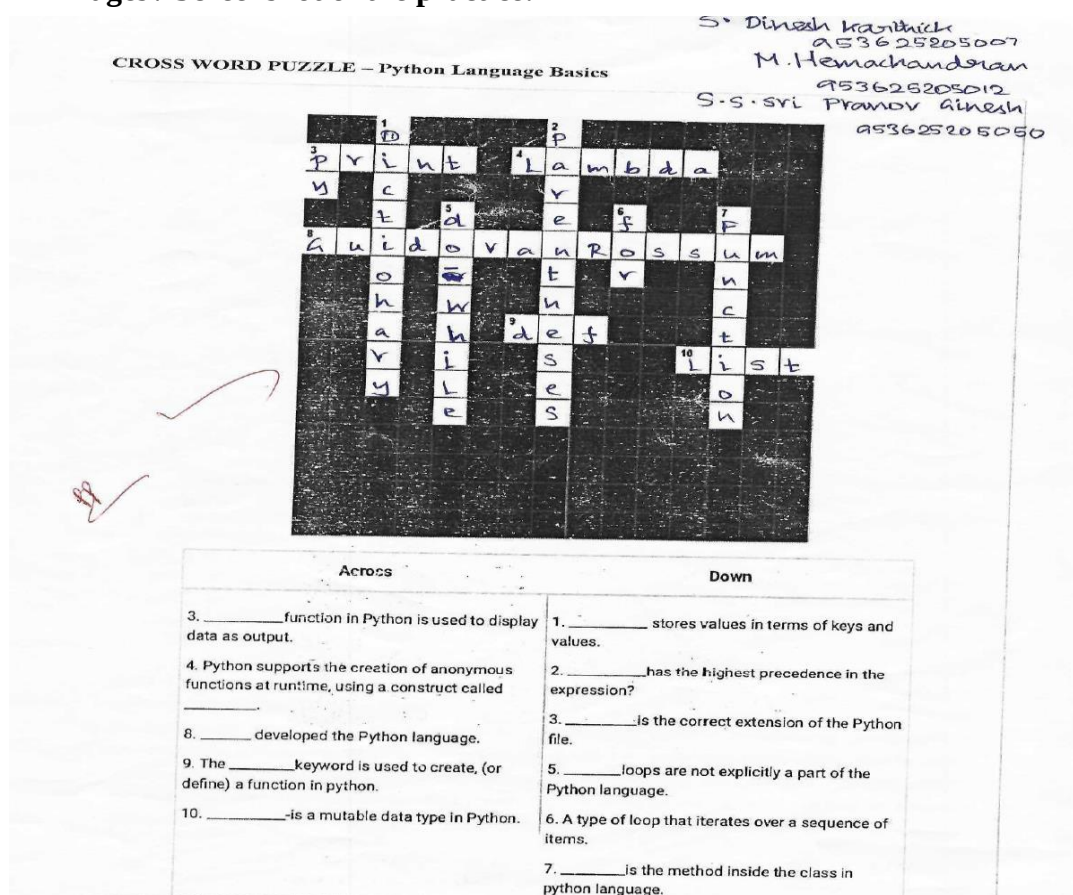


Figure 1. a Puzzle paper

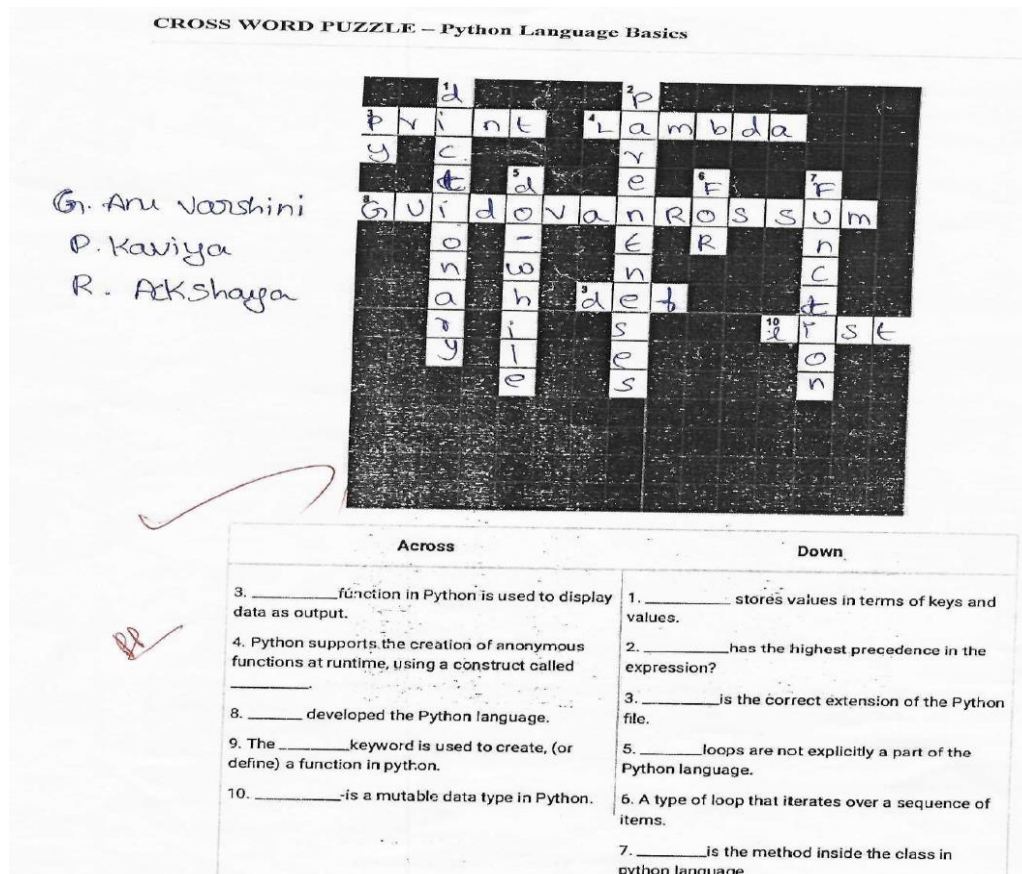


Figure 1. b puzzle paper

Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

- A large majority of students participated enthusiastically in the crossword activity, completing the puzzles readily and showing a positive response to the revision method.
- Students demonstrated strong proficiency in recognizing straightforward, definition-based terminology like "syntax," "parameters," and simple definitions.
- Conversely, students struggled significantly to identify clues that related to the more abstract logical and conceptual aspects of Python programming.

❖ Benefit of the practice:

- Students treated the crossword puzzle as an opportunity to reinforce terminology and improve their understanding of **python** programming fundamentals
- Students shared programming knowledge and collaborated on solving clues with their peer students, thereby enhancing teamwork and communication skills.
- This practice improves the analytical and critical thinking skills of the students as they

logically deduce terms from contextual clues.

❖ ***Challenges faced in implementation:***

- Students perceived the crossword puzzle as a valuable opportunity to solidify their understanding of Python programming fundamentals and reinforce key terminology.
- The activity fostered peer-to-peer knowledge sharing and collaboration, allowing students to work together on solving clues and thereby enhancing their teamwork and communication skills.
- This practice significantly contributed to the improvement of students' analytical and critical thinking abilities, as they were required to use logical deduction to determine terms from contextual clues.

References:

<https://www.geeksforgeeks.org/c/a-c-puzzle/>

https://wordmint.com/public_puzzles/1248202

Signature of Faculty Member

HOD